Swiggy SQL Data Analysis

Advanced SQL queries for strategic insights from food delivery data

Project Overview

Dataset

Customers, restaurants, orders, delivery partners, and ratings

Goal

Extract insights for business strategy and operational efficiency

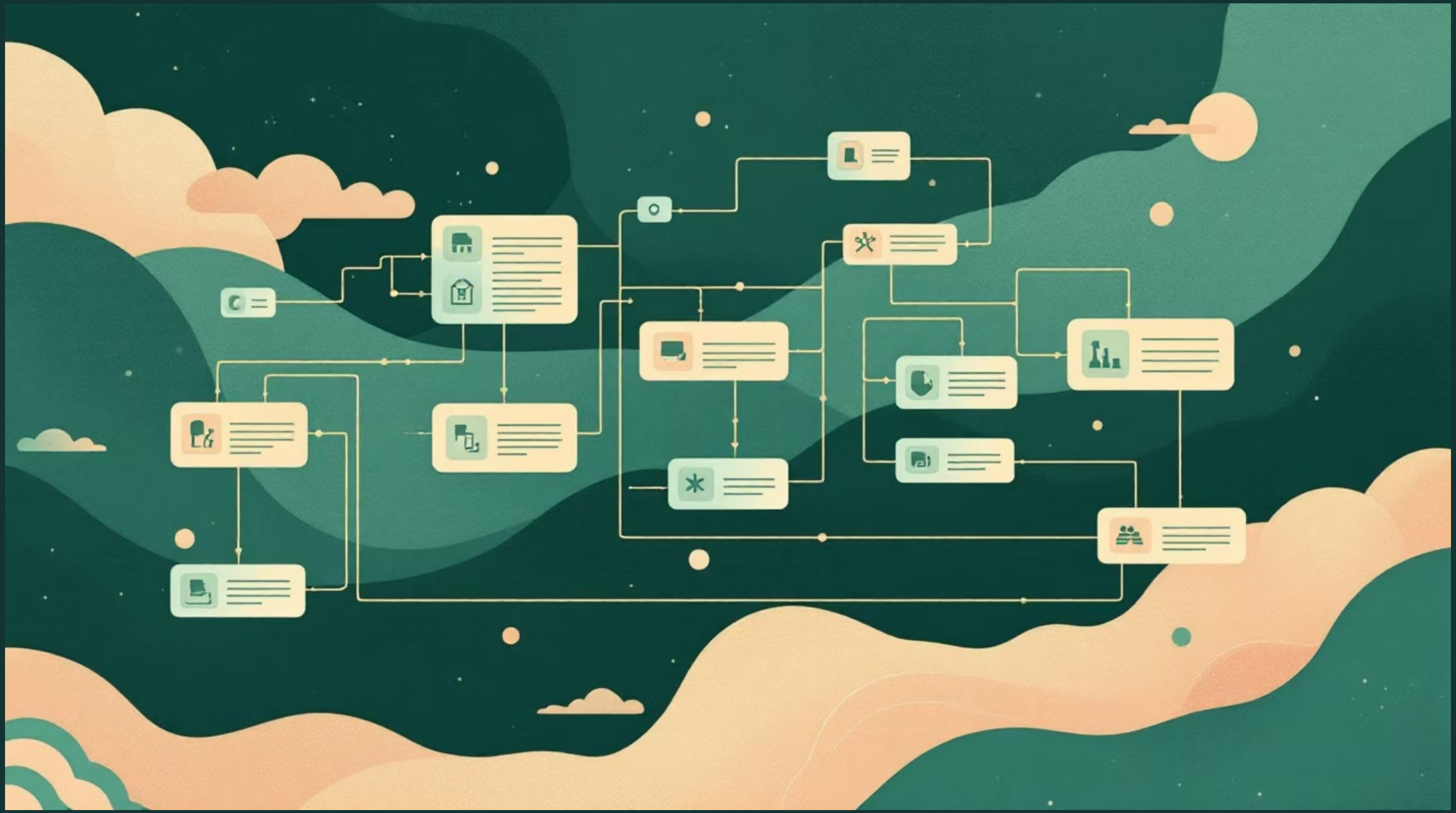
Technologies

SQL with joins, aggregations, subqueries, and date functions

Database Schema

Core Tables

- Customers (customer_id, name, city)
- Restaurants (restaurant_id, name, city, rating)
- Orders (order_id, customer_id, restaurant_id, order_date, amount)
- Delivery Partners (partner_id, name)
- Deliveries (delivery_id, order_id, partner_id)



Customer Analysis

01

Delhi Residents

SELECT all customers where city = 'Delhi'

02

Active Customers

JOIN customers with orders to find those who placed at least one order

03

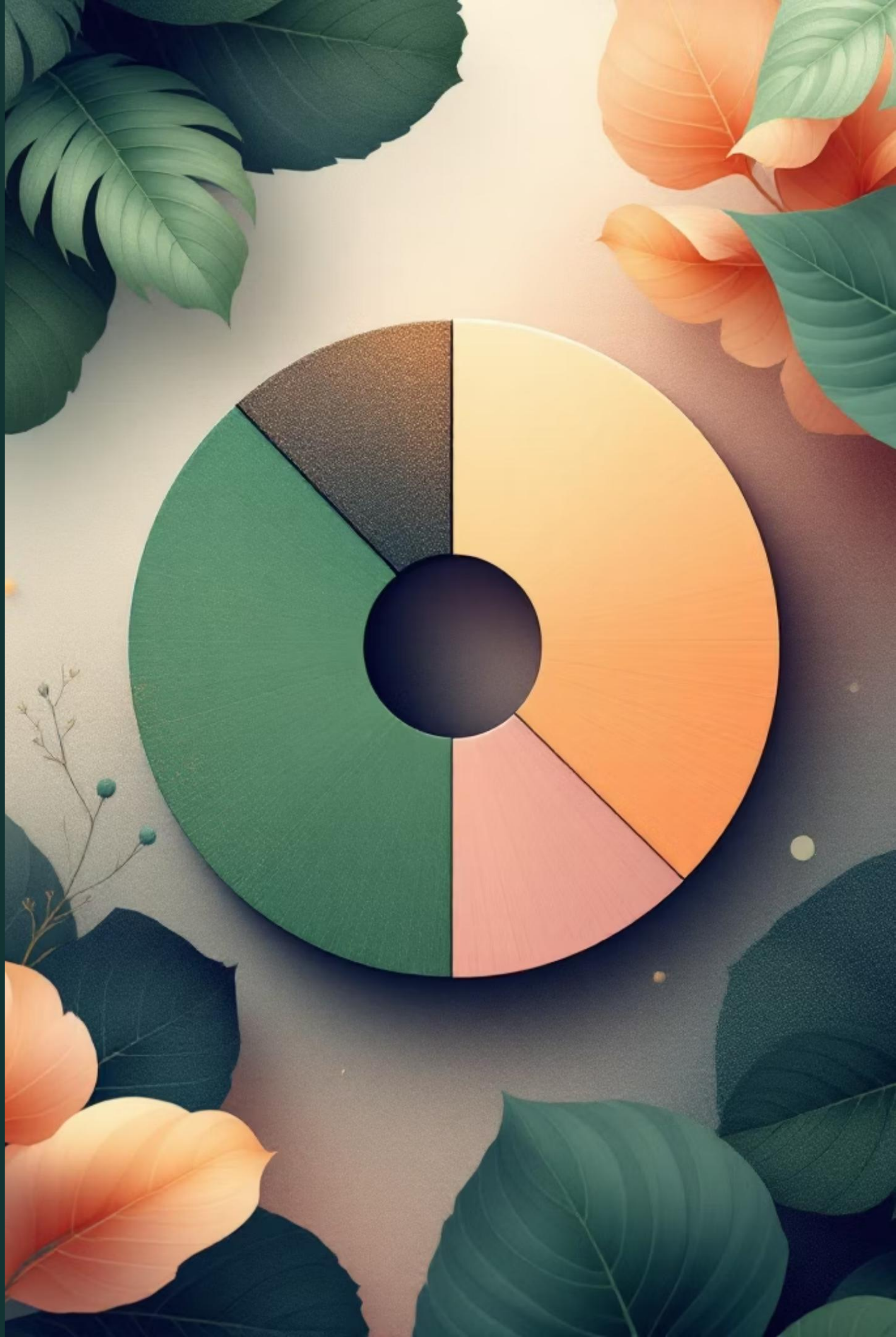
Order Counts

GROUP BY customer_id with COUNT to get total orders per customer

04

Inactive Customers

LEFT JOIN with NULL check to identify customers who never ordered



Restaurant Performance

1

Average Mumbai Ratings

AVG(rating) FROM restaurants WHERE city = 'Mumbai'

2

Top 5 Restaurants

ORDER BY rating DESC LIMIT 5 to find highest-rated establishments

3

Revenue Analysis

SUM(amount) GROUP BY restaurant to calculate total revenue generated

Delivery Operations

Delivery Partner Metrics

Query identifies partners who completed more than 1 delivery using GROUP BY and HAVING clause.

Most Connected Partner

Finds delivery partner serving most unique customers through multiple JOIN operations and DISTINCT counting.

Temporal Analysis

1. ## Recent Orders

Orders in last 30 days using CURRENT_DATE - INTERVAL 30 DAY

2. ## 3-Day Pattern

Customers ordering on exactly 3 different days with COUNT(DISTINCT order_date)

Key Insights

Customer Engagement

Significant variation in ordering patterns across different cities

Rating vs Revenue

High-rated restaurants don't always generate highest revenue

Delivery Concentration

Small number of partners handle large share of deliveries

Repeat Customers

Loyal customers contribute significantly to total order volume

Strategic Applications

Customer Segmentation

Identify active vs inactive users for targeted marketing

Revenue Analysis

Track performance by restaurant and location

Operational Optimization

Improve delivery efficiency and partner allocation

Future Improvements

BI Integration

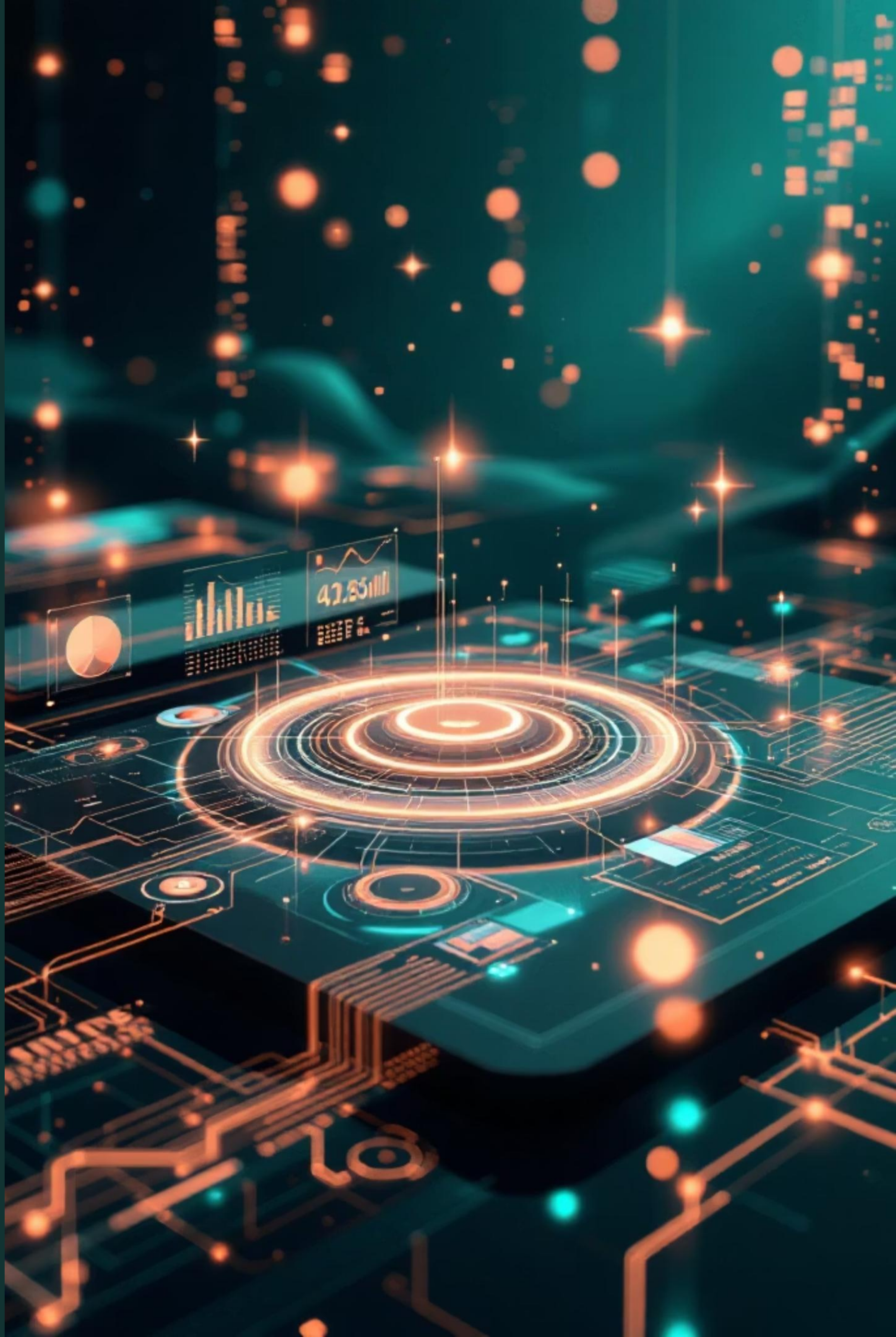
Connect with Power BI or Tableau for interactive dashboards

Predictive Analytics

Add customer churn prediction and demand forecasting

Query Optimization

Enhance performance for large-scale datasets with advanced indexing



DATA LINK

https://drive.google.com/drive/folders/1Hza0MPqeYwUuY7f77IykVtcp9gQwkUPz?usp=drive_link

GITHUB

https://github.com/Rakesh5803/05-Swiggy.git

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